SCRIPT

You are being invited to participate, as a volunteer, in the research study **“Perceptions of Data Quality Requirements for Machine Learning-Based Systems”**, conducted by **<Anonymized>** and **<Anonymized>**.

In this study, we seek to understand how developers treat data quality as a requirement in machine learning-based systems. Our purpose is to identify existing challenges and gaps in this area. To this end, we invite you to collaborate with our research by answering the following questions.

To participate in this study, you must be over 18 years old and have previously worked in software development for machine learning-based systems. Your participation consists of answering an online questionnaire, which will take approximately 15 minutes to complete. We guarantee the secrecy and confidentiality of all information provided during the study.

The Informed Consent Form aims to ensure your rights as a participant and is available at this **<link>**. Please read it carefully to fully understand the purpose of the research. Your participation is voluntary, and you may choose not to participate or withdraw your consent at any time, without any penalty or harm.

After receiving clarification about the nature of the research, its objectives, and methods, I declare that I am over 18 years old and agree to participate through the participation form, in accordance with the provisions of the Brazilian General Data Protection Law (Law No. 13,709/2018). In the event of a negative response, the form will be closed, thereby ensuring respect for the autonomy and privacy of participants, as established by the legislation in force. **(Required)**

( ) I agree to participate
( ) I do not agree to participate

GENERAL CHARACTERIZATION OF THE PARTICIPANT

In this section, we will collect basic information and personal data to better understand your profile.

1) How old are you? (Required)

( ) 18-24 years old
( ) 25-34 years old
( ) 35-44 years old
( ) 45-54 years old
( ) 55-64 years old
( ) 65 years or older.
( ) I prefer not to say.

2) In which country, state do you currently reside? (Required)

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3) What is your gender? (Required)

( ) Man
( ) Woman
( ) Non-binary, genderqueer, or gender non-conforming
( ) Prefer not to say
( ) Other

4) What is your level of education? (Please choose one) (Required)

( ) High School
( ) Higher education
( ) Specialization
( ) Master's student
( ) Master
( ) Doctoral student
( ) Doctorate

5) Which role best describes your current activities in the development of machine learning-enabled systems? (Please choose one) (Required)

( ) Software Developer (Backend, front-end, fullstack)
( ) Data engineer
( ) Data scientist
( ) DevOps engineer
( ) Product owner
( ) Tech manager
( )QA engineer
( ) Other:____________

6) **How senior is your role in the team? (Please choose one) (Required)**
   ( ) Trainee
   ( ) Junior (up to 5 years)
   ( ) Full (6 to 9 years)
   ( ) Senior (10+ years)

7) **How many machine learning-based projects have you been involved in? Please provide your best estimate. (Required)**

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CHARACTERIZATION OF DATA QUALITY EXPERIENCE IN MACHINE LEARNING-ENABLED SYSTEMS

In this section, we would like to learn about your experience with data quality in machine learning-based systems.

8) **How would you rate your skill level in relation to the following activities that involve the data processing stage in a machine learning-enabled system? (Required)**

Skill	**Very high**	**Above average**	**Averag e**	**Below Average**	**Very low**	**Not applicab le**
Data cleaning (removing duplicates, filling in missing values, etc.)						
Normalizing or standardizing data (adjusting variable scales, transforming categorical variables, etc.)						
Detecting and removing outliers						
Integrating multiple data sources (joining different datasets, resolving						

inconsistencies between sources)						
Application of data transformation techniques (e.g., PCA, coding of categorical variables)						
Data validation (consistency, accuracy and completeness checks)						
Automating data processing pipelines						
Monitoring the quality of data in production (creating alerts and metrics for continuous quality)						
Use of libraries and tools for data processing (e.g. Pandas, PySpark, Dask)						
Splitting data into training and testing						

PERCEPTION OF DATA QUALITY AS A REQUIREMENT FOR MACHINE LEARNING-ENABLED SYSTEMS

In this section, we seek to understand their perspectives on different characteristics of data quality for machine learning-enabled systems

Data quality is a fundamental aspect of successful machine learning projects. Poor quality data can compromise the accuracy, reliability and generalizability of trained models, affecting the expected results.

9) **To explore your perception of this topic, what are the first five words that come to mind when you think of data quality as a requirement in machine learning projects? (Required)**

   Word 01: ________________

Word 02: ________________
Word 03: ________________
Word 04: ________________
Word 05: ________________

10) **What experience do you have with Requirements Engineering activities in your projects? Have you ever been involved in defining, analyzing or documenting requirements? If so, can you share a little about what that process was like? (Required)**

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To answer the next questions, consider the description of the following **data quality characteristics:**

- **Accuracy:** This refers to the degree to which the data is correct, accurate and error-free. This implies that the data values faithfully represent reality or the characteristics they are intended to measure or describe. Accurate data avoids errors that can compromise the results of machine learning models or analyses.
- **Completeness:** A measure of how many expected values are present in a data set. Complete data contains all the necessary information, with no gaps that could affect analysis or model training.
- **Consistency:** Indicates that the data has no contradictions within or between different sources. Consistency is essential to ensure that the same variables maintain the same values and meanings throughout the system.
- **Credibility:** Refers to trust in the data source. Credibility is important for the users or systems that consume the data to trust it when making decisions.
- **Timeliness:** Reflects how recent and relevant the data is in relation to when it is used. Up-to-date data is critical in areas where conditions change rapidly, such as demand forecasting or financial analysis.
- **Accessibility:** Refers to the ease with which data can be obtained and used by those who need it, ensuring that it is available at the right time, in the appropriate format available and with the necessary access rights.
- **Compliance:** This refers to the alignment of data with established norms, regulations or standards. For example, health data needs to comply with privacy laws such as the LGPD or GDPR.
- **Reliability:** Measuring the ability of data to be used consistently and stably over time. Reliable data is data that repeatedly produces predictable and correct results.
- **Efficiency:** Indicates the ability of data to be processed and used effectively, without overloading systems. Efficient data is structured in a way that optimizes the use of computing resources.
- **Traceability:** Refers to the ability to follow the history of data, from its origin to the transformations that have taken place over time. This is important for auditing and troubleshooting.

- **Comprehensibility:** Evaluates how easily the data can be interpreted and understood by users or systems. Comprehensible data has clear documentation and standardized structures.
- **Availability:** This refers to the ability to ensure that data is accessible and usable whenever it is needed, without unexpected interruptions or restrictions. This characteristic is essential if systems and projects are to operate continuously and reliably.
- **Recoverability:** This refers to the ability to restore data after a failure or loss. Recoverable data guarantees continuity of operations in the event of incidents such as hardware failures or human error.

11) Based on your experience, how important are the following data quality characteristics for machine learning-enabled systems? (Required)

Characteristic	**Not important at all**	**Not very important**	**Neutral**	**Important**	**Very important**
Accuracy					
Completeness					
Consistency					
Credibility					
Timeliness					
Accessibility					
Compliance					
Reliability					
Efficiency					
Traceability					
Comprehensibility					
Availability					
Recoverability					

12) **For those characteristics that you considered important, please describe the reason for your rating. (Optional)**

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13) **Regarding data quality characteristics, how would you characterise the degree of priority of each characteristic for machine learning-based system projects? (Required)**

Characteristic	**Not a priority**	**Low priority**	**Neutral**	**High priority**	**Essential**
Accuracy					
Completeness					
Consistency					
Credibility					
Timeliness					
Accessibility					
Compliance					
Reliability					
Efficiency					
Traceability					
Comprehensibility					
Availability					
Recoverability					

14) **For those characteristics that you considered to be a priority, please describe the reason for your classification. (Optional)**

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15) **Based on your experience, how do you balance different data quality characteristics, such as accuracy, comprehensibility and computational**

efficiency, when developing machine learning-enabled systems? Describe situations and decisions in which different quality characteristics had to be reconciled. (Required)

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16) **How can data version control impact data quality characteristics in machine learning projects? (Please choose one) (Required)**

( ) Ensures data consistency and traceability over time, allowing changes to the dataset to be documented and verified.
( ) Increases the amount of data available, without the need to check consistency between versions.
( ) Eliminates the need to document changes to the dataset, as all changes are automatically applied to the model.
( ) Reduces data accuracy, as old versions are no longer used in models.
( ) Other:____________

IMPLEMENTATION OF DATA QUALITY REQUIREMENTS IN THE DEVELOPMENT PROCESS

In this section, we seek to understand how data quality requirements are taken into account in the process of developing machine learning-enabled systems.

17) **Considering the machine learning projects you have participated in, how is data quality incorporated into the development process? (Multiple choices possible) (Required)**

[ ] Initial assessment during data collection and preparation
[ ] Continuous monitoring throughout the model's life cycle
[ ] Test sets are applied to validate the consistency, completeness and accuracy of the data before it is used for training.
[ ] There is no formal strategy for ensuring data quality during development.
[ ] Other:___________

18) **How do you measure the impact of data quality on the performance of the machine learning model? (Multiple choices possible) (Required)**

[ ] A/B testing
[ ] Analysis of performance metrics (e.g. precision, recall)
[ ] Manual review of results
[ ] Other:____________

19) **How often are data quality requirements formally discussed during the development of machine learning projects? (Please choose one) (Required)**

( ) Every day
( ) At least once a week, but not every day
( ) Less than once a week, but at least once a month
( ) Less than once a month
( ) Never

20) **How do you document and communicate data quality issues within your team and with other stakeholders? (Multiple choices possible) (Required)**

[ ] Structured language (text)
[ ] Project management tools (Jira, Trello or Asana)
[ ] Centralized documentation (systems such as Confluence, Google Docs or Notion)
[ ] Alignment meetings
[ ] Periodic reports
[ ] Other:____________

21) **What are the main challenges in guaranteeing data reliability in machine learning projects? (Multiple choices possible) (Required)**

[ ] Inconsistency between different data sources
[ ] Incomplete or missing data
[ ] Lack of standardization in data formats
[ ] Outdated or unreliable data
[ ] Errors introduced during collection and processing
[ ] Difficulty in data traceability and versioning
[ ] Lack of adequate tools for validating data quality
[ ] Other:____________

22) **How often do you receive support from other teams (e.g. data team, data scientists) to deal with data quality issues? (Please choose one) (Required)**

( ) Always
( ) Often
( ) Occasionally
( ) Rarely
( ) Never